STUDENT PERFORMANCE GRADE PREDICTION

A Machine Learning Approach

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PROJECT OVERVIEW

CORE IDEA

- Predict student final grades using Machine Learning
- Identify at-risk students early
- Support academic intervention

OBJECTIVES

- Predict grades accurately
- Compare ML models
- Select best model
- Improve decision-making

DATASET OVERVIEW

10,000
TOTAL STUDENTS

9,786
CLEAN RECORDS

12
TOTAL FEATURES

DEMOGRAPHIC

- Gender
- Ethnicity
- Parental Education

ACADEMIC

- Math Score
- Reading Score
- Writing Score
- Science Score

SUPPORT

- Lunch Type
- Test Preparation

GRADE LABELING & PROBLEM TYPE

GRADE	SCORE RANGE
A	90 – 100
B	80 – 89
C	70 – 79
D	60 – 69
Fail	Below 60

PROBLEM DEFINITION

MULTI-CLASS CLASSIFICATION

Categorical Labels

DATA CLEANING

PREPROCESSING STEPS

- Removed outliers using the **IQR Method** (Interquartile Range)
- Handled missing values to ensure data integrity
- Improved dataset quality and reliability for modeling

FINAL DATASET SIZE

9,786

Cleaned Student Records

FEATURE ENGINEERING

ENCODING

- Label Encoding for categorical variables
- Converts text data to numerical format

SCALING

- StandardScaler normalization

STANDARDIZATION PARAMETERS

$\mu = 0$

MEAN

$\sigma = 1$

VARIANCE

IMPORTANCE

- Improves model performance & convergence
- Critical for distance-based models (KNN)

TRAIN-TEST SPLIT

80% TRAINING

20% TESTING

METHODOLOGY

- Stratified sampling used to maintain class proportions
- Ensures the model is evaluated on a representative sample
- Prevents bias in grade distribution across sets

✓ **ENSURES BALANCED CLASS DISTRIBUTION**

MACHINE LEARNING MODELS

DECISION TREE

Simple rule-based classification model

RANDOM FOREST

Ensemble learning algorithm using multiple trees

K-NEAREST NEIGHBORS

Similarity and distance-based classifier (K=5)

MODEL EXPLANATION

RULE-BASED

DECISION TREE

- Flowchart-like structure for decision making
- Fast training and easy to interpret
- High risk of overfitting on complex data

ENSEMBLE LEARNING

RANDOM FOREST

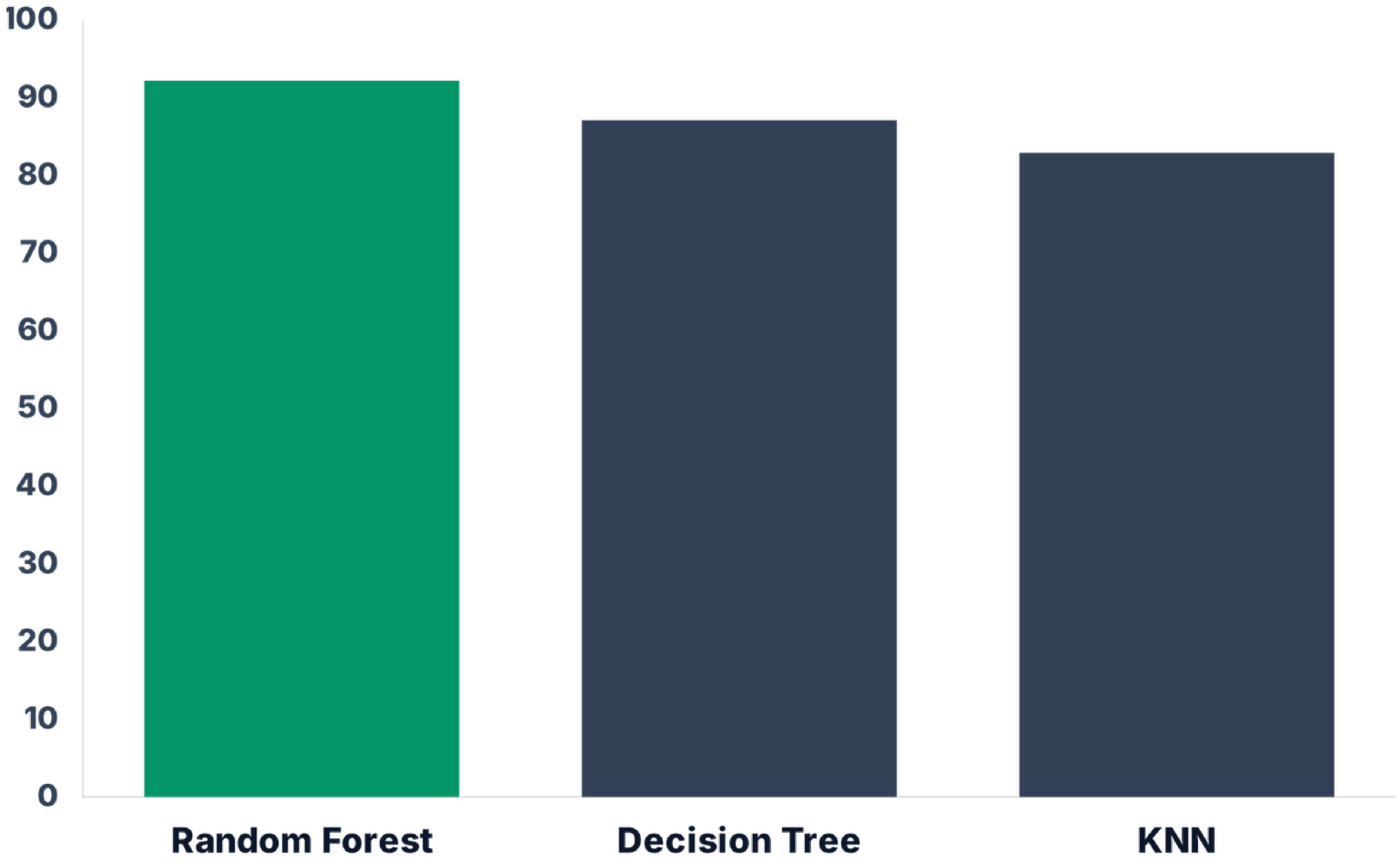
- Collection of multiple decision trees
- Highly accurate and reduces variance
- Stable performance across datasets

INSTANCE-BASED

K-NEAREST NEIGHBORS

- Classifies based on feature similarity
- Simple implementation (K=5 used)
- Computationally slower with large data

MODEL PERFORMANCE



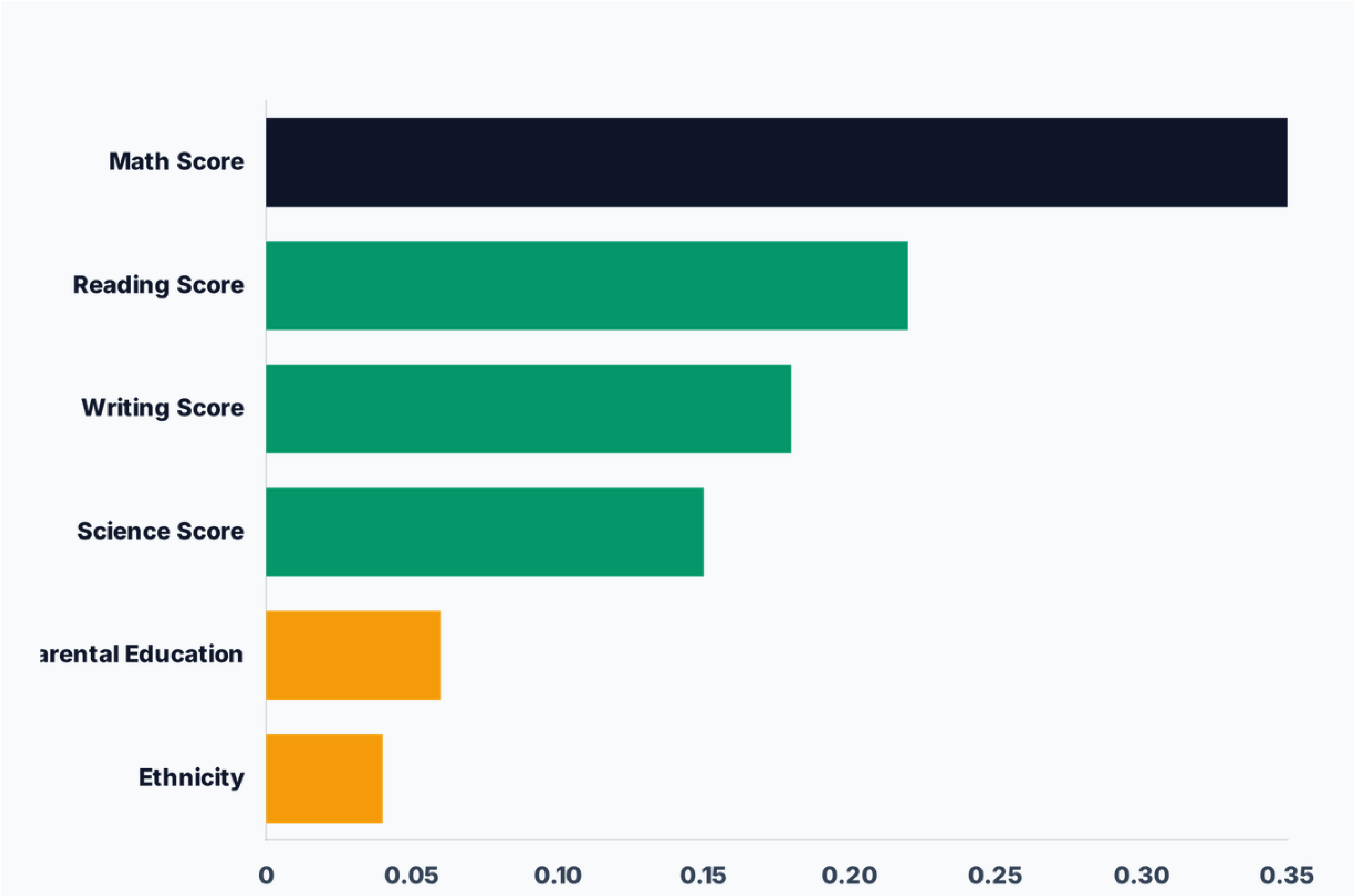
MODEL	ACCURACY
Random Forest	91.82%
Decision Tree	87.48%
KNN (K=5)	83.24%

✓ RANDOM FOREST IS THE BEST MODEL

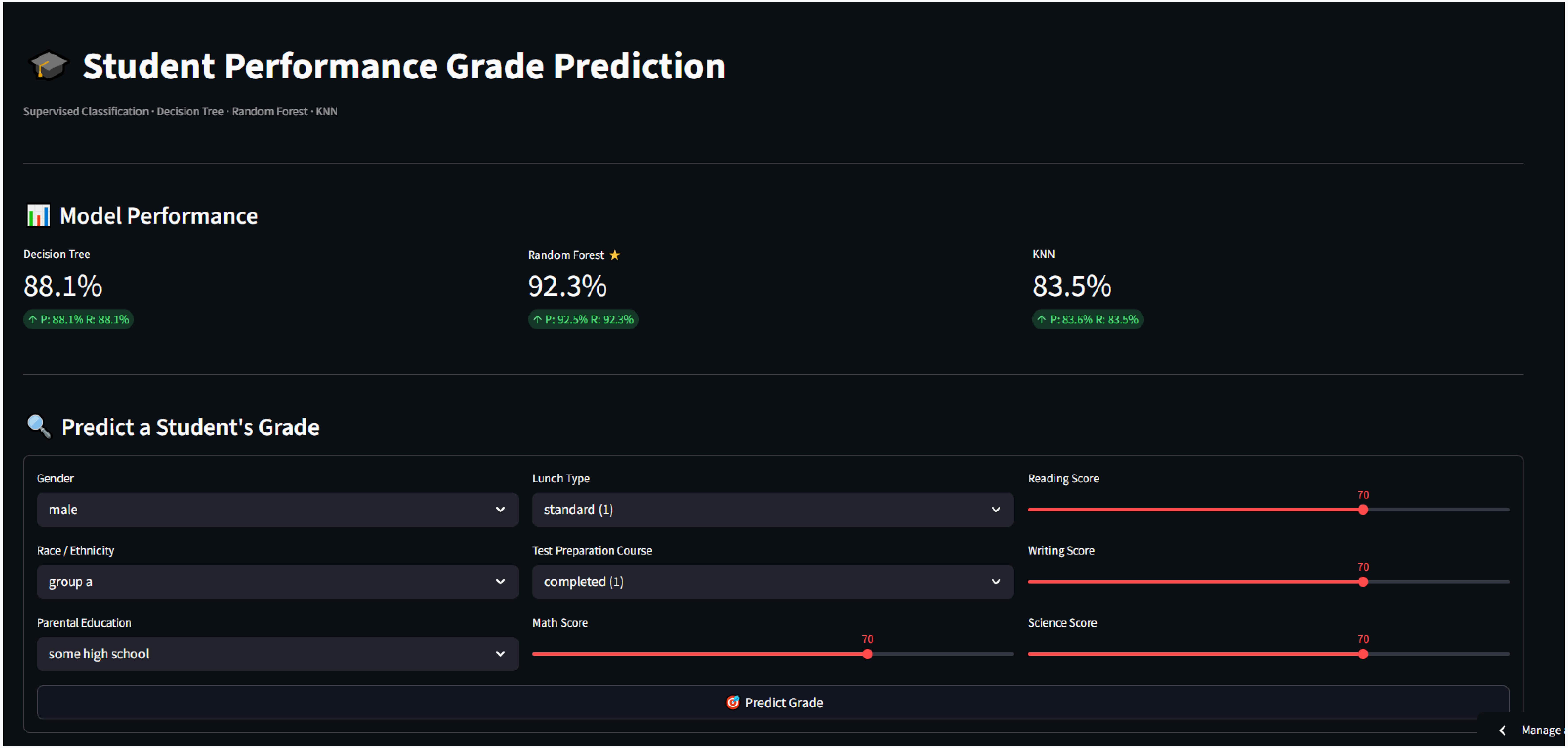
FEATURE IMPORTANCE

KEY FINDING

- Academic scores are the strongest predictors of final grades.
- Math score shows the highest correlation with performance.



GUI


Student Performance Grade Prediction
Supervised Classification · Decision Tree · Random Forest · KNN
Model Performance
Decision Tree
88.1%
P: 88.1% R: 88.1%
Random Forest
92.3%
P: 92.5% R: 92.3%
KNN
83.5%
P: 83.6% R: 83.5%
Predict a Student's Grade
Gender
male
Lunch Type
standard (1)
Reading Score
70
Race / Ethnicity
group a
Test Preparation Course
completed (1)
Writing Score
70
Parental Education
some high school
Math Score
70
Science Score
70
Predict Grade
Manage

CONCLUSION

- **Random Forest** achieved the best predictive performance (92.19% Accuracy)
- **Academic scores** (Math, Reading, Writing) were the strongest predictors
- Machine Learning effectively supports **early student intervention**

PRACTICAL IMPACT

Supports better educational decision-making and identifies struggling students earlier.

FUTURE WORK

Incorporate attendance and study-hour features

Build a real-time academic dashboard

Explore Deep Learning models (ANN)

THANK YOU